IGCSE 0607 Practice Questions — E1.11 — Combined Questions & Markscheme

Rates

Total Questions: 21 — **Total Marks:** 114 Marks

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- **E1.11** Rates
  → **Paper 2 (Non-Calculator)** *(7 Qs)* · **Paper 4 (Calculator)** *(14 Qs)*

1 A train departs at 08 : 15 and arrives at its destination at 09 : 45.
The train travels a distance of 120 km.
Calculate the average speed of the train.

..... km/h **[2]**

2 A train travels 102 km in 1 hour 12 minutes.
Work out its average speed.

..... km/h **[3]**

3 A car travels 120 km in 1 hour and 30 minutes.

Calculate the average speed of the car in km/h.

..... km/h **[3]**

4 **(a)** A car travels a distance of 108 km in 1 hour 30 minutes.
Calculate its average speed in km/h.

..... km/h **[2]**

(b) Change your answer in part (a) into m/s.

..... m/s **[2]**

5 **(a)** A train travels a distance of 120 km in 1 hour and 30 minutes.
Calculate the average speed of the train.

..... km/h **[2]**

(b) A car uses 8 litres of fuel for every 100 km travelled.
Work out how much fuel the car uses to travel 350 km.

..... litres **[2]**

6 **(a)** A train travels a distance of 180 km in 2 hours and 15 minutes.
Calculate the average speed of the train in km/h.

..... km/h **[2]**

(b) An empty tank is filled with water at a constant rate of 1.5 litres per second.
Work out the time it takes to fill the tank with 450 litres of water.
Give your answer in minutes.

..... minutes **[2]**

7 A cyclist travels 30 km at an average speed of v km/h.
He then travels a further 40 km at an average speed of $(v + 5)$ km/h.
The total time taken for the entire journey is 4 hours.

(a) Write down an equation in terms of v and show that it simplifies to $2v^2 - 25v - 75 = 0$.

[3]

(b) Solve the equation $2v^2 - 25v - 75 = 0$ to find the possible values of v.

$v =$ ..... , or $v =$ ..... **[3]**

(c) Find the time taken for the first 30 km of the journey.

..... hours **[1]**

8 A train travels a distance of 385 km.
The journey takes 4 hours 25 minutes.
Calculate the average speed of the train in km/h.
Give your answer correct to 1 decimal place.

..... km/h **[3]**

9 Water flows from a pipe into a rectangular tank at a rate of 45 litres per minute.

(a) Calculate the volume of water, in cubic centimetres (cm^3), that flows into the tank in 1 second.

..... cm^3 **[2]**

(b) The tank has a base measuring 1.5 m by 80 cm. Calculate the rate at which the water level is rising in the tank, giving your answer in cm/s.

..... cm/s **[3]**

10 A car uses fuel at an average rate of 6.4 litres per 100 km.

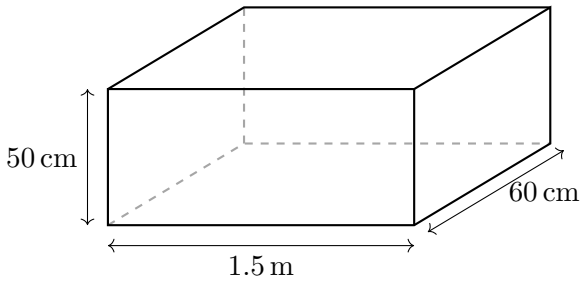
(a) Calculate the amount of fuel used for a journey of 350 km.

..... litres **[2]**

(b) The cost of fuel is 1.35 euros per litre.
The exchange rate is \$1 = 0.85 euros. Calculate the cost of the fuel for this journey in dollars (\$).
Give your answer correct to the nearest cent.

\$ ..... **[3]**

11 A rectangular water tank measures 1.5 m by 60 cm by 50 cm.



(a) Calculate the volume of the tank in litres.

..... litres **[2]**

(b) Water is pumped into the empty tank at a constant rate of 24 litres per minute.
Calculate the time taken to fill the tank completely.
Give your answer in minutes and seconds.

minutes ..... , seconds ..... **[3]**

12 A swimming pool has a volume of 52 m^3.

(a) Water flows into the empty pool from a pipe at a rate of 1.6 litres per second.
Calculate the time taken to completely fill the pool.
Give your answer in hours and minutes, correct to the nearest minute.

hours, minutes **[3]**

(b) The cost of the water is \$2.15 per cubic metre. The exchange rate is \$1 = 0.84 euros.
Calculate the total cost of the water used to fill the pool in euros.
Give your answer correct to 2 decimal places.

euros ..... **[3]**

13 **(a)** A car travels 325 km.
It uses fuel at a rate of 8.4 litres per 100 km.
Fuel costs \$1.35 per litre.
Calculate the total cost of the fuel used for this journey.

\$ ..... **[3]**

(b) On a different journey, the car travels at an average speed of 75 km/h.
The journey takes 4 hours and 24 minutes.
Work out the distance travelled.

..... km **[3]**

14 Water flows from a pipe into a rectangular tank at a rate of 2.5 litres per second.
The tank has a length of 1.8 m, a width of 1.5 m, and a height of 1.2 m.
The tank is initially empty.

(a) Calculate the capacity of the tank in litres.

..... litres **[2]**

(b) Work out how long it takes to fill the tank completely.
Give your answer in minutes.

..... minutes **[2]**

(c) Another pipe empties the full tank at a rate of x litres per minute.
It takes 45 minutes to empty the tank completely.
Calculate the value of x.

$x =$ ..... **[2]**

15 **(a)** Water flows from a pipe into an empty tank at a rate of 0.35 litres per second.
The tank is a cuboid with length 2.5 m, width 1.4 m and height 1.2 m.

Calculate the time taken to fill the tank completely.
Give your answer in hours and minutes.

h ..... , min ..... **[4]**

(b) A mobile phone costs \$850 in the USA and €780 in Germany. The exchange rate is \$1 = €0.84.

Calculate the difference in price between the two countries.
Give your answer in dollars (\$).

\$ ..... **[3]**

16 A car travels from City A to City B, a distance of d km, at an average speed of v km/h.
The car then travels from City B to City C, a distance of $2d$ km, at an average speed of $(v + 15)$ km/h.
The total time taken for the entire journey from City A to City C is T hours.

(a) Show that the total time T can be written as

$$T = \frac{3dv + 15d}{v(v + 15)}$$

[3]

(b) Given that $d = 120$ and $T = 4.5$,

Form an equation in v and show that it simplifies to

$$3v^2 - 195v - 1200 = 0$$

[4]

17 A cyclist travels x km at a speed of 12 km/h. She then travels a further $(x + 5)$ km at a speed of 15 km/h. The total time taken for the whole journey is 2 hours 35 minutes.

(a) Work out the value of x.

$x =$ ..... **[4]**

(b) Calculate her average speed for the whole journey.
Give your answer correct to 1 decimal place.

..... km/h **[3]**

18 An aeroplane flies from City A to City B, a distance of 2400 km.

(a) The aeroplane flies at an average speed of v km/h.

Write down an expression, in terms of v, for the time taken in hours.

..... hours **[1]**

(b) The aeroplane returns from City B to City A.
The average speed on the return journey is $(v + 40)$ km/h.
The total time taken for the two flights is 10 hours and 30 minutes.
Write down an equation in v and show that it simplifies to

$$7v^2 - 2920v - 192000 = 0$$

[4]

(c) Solve the equation $7v^2 - 2920v - 192000 = 0$ to find the average speed of the aeroplane from City A to City B.
Give your answer correct to the nearest integer.

..... km/h **[2]**

19 A cyclist travels from Town A to Town B, a distance of 48 km.
The cyclist travels the first 30 km at an average speed of x km/h. They travel the remaining distance at an average speed of $(x - 4)$ km/h. The total time taken for the journey is 2 hours and 30 minutes.

(a) Show that $5x^2 - 116x + 240 = 0$.

[4]

(b) Solve the equation $5x^2 - 116x + 240 = 0$ to find the possible values of x.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... **[3]**

20 A man runs 15 km at an average speed of x km/h.
He then walks 12 km at an average speed of $(x - 4)$ km/h.
The total time taken for the journey is 3.5 hours.

(a) Show that x satisfies the equation

$$7x^2 - 82x + 120 = 0$$

[3]

(b) **(i)** Solve the equation

$$7x^2 - 82x + 120 = 0$$

$x =$ ..... , or $x =$ ..... **[3]**

(ii) Find the time taken for the man to run the 15 km.

..... hours **[2]**

21 A car travels 150 km at an average speed of x km/h.

(a) Write down an expression, in terms of x, for the time taken in hours.

..... hours **[1]**

(b) The car then travels a further 90 km at an average speed of $(x + 15)$ km/h.
The total time for the whole journey is 3 hours.
Show that $x^2 - 65x - 750 = 0$.

[4]

(c) Solve the equation $x^2 - 65x - 750 = 0$ to find the initial average speed of the car.

$x =$ ..... km/h **[3]**

1 Time taken = 1.5 hours (or 1 hour 30 minutes) **[B1]**

$120 \div 1.5 = 80$ **[B1]**

[Back to Question] *[Total Marks: 2]*

2 1 hour 12 minutes = 1.2 hours or 72 minutes **[B1]**

$\frac{102}{1.2}$ or $\frac{102}{72} \times 60$ **[M1]**

85 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 Time = 1.5 hours **[B1]**

Average Speed = $\frac{120}{1.5}$ **[M1]**

80 **[A1]**

[Back to Question] *[Total Marks: 3]*

4 (a) M1 for $108 \div 1.5$ or $\frac{108}{90} \times 60$

72 **[A1]**

(b) M1 for $72 \times \frac{1000}{3600}$ or $72 \div 3.6$

20 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) $\frac{120}{1.5}$ or $\frac{120}{90} \times 60$ **[M1]**

80 **[A1]**

(b) $\frac{8}{100} \times 350$ or 8×3.5 **[M1]**

28 **[A1]**

[Back to Question] *[Total Marks: 4]*

6 (a) Time = 2.25 or $\frac{9}{4}$ hours **[M1]**

$180 \div 2.25 = 80$ **[A1]**

(b) Time in seconds = $450 \div 1.5 = 300$ **[M1]**

$300 \div 60 = 5$ **[A1]**

[Back to Question] *[Total Marks: 4]*

7 **(a)** M1 for $\frac{30}{v} + \frac{40}{v+5} = 4$

M1 for $30(v+5) + 40v = 4v(v+5)$ (removing fractions)

A1 for obtaining $2v^2 - 25v - 75 = 0$ with no errors or omitted steps in working

(b) M2 for $(2v+5)(v-15) = 0$ or correct substitution into the quadratic formula
(M1 for $(2v+a)(v+b) = 0$ where $ab = -75$ or $2b + a = -25$)

$v = 15$ and $v = -2.5$ **[A1]**

(c) 2 **[B1]**

[Back to Question] *[Total Marks: 7]*

8 Time $= 4\frac{25}{60}$ hours or 4.416... hours **[M1]**

Average Speed $= \frac{385}{4.416...}$ **[M1]**

87.2 (correct to 1 decimal place) **[A1]**

[Back to Question] *[Total Marks: 3]*

9 **(a)** 45×1000 or $\frac{45}{60}$ **[M1]**

$\frac{45000}{60} = 750$ **[A1]**

(b) Base area $= 150 \times 80 = 12000$ cm^2 **[M1]**

Rate of height increase $= \frac{\text{Volume per second}}{\text{Base area}} = \frac{750}{12000}$ **[M1]**

0.0625 **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** $\frac{350}{100} \times 6.4$ **[M1]**

22.4 **[A1]**

(b) Cost in euros $= 22.4 \times 1.35 = 30.24$ **[M1]**

Cost in dollars $= 30.24 \div 0.85$ **[M1]**

35.58 **[A1]**

[Back to Question] *[Total Marks: 5]*

11 **(a)** $150 \times 60 \times 50$ or 450000 seen or $1.5 \times 0.6 \times 0.5 = 0.45$ m^3 **[M1]**

450 **[A1]**

(b) $\frac{450}{24}$ or 18.75 **[M1]**

0.75×60 or 45 **[M1]**

18 minutes 45 seconds **[A1]**

[Back to Question] *[Total Marks: 5]*

12 **(a)** M1 for 52×1000 or 52000 seen

M1 for their $52000 \div 1.6$ (implies 32500 seconds)

9 hours 2 minutes **[A1]**

(b) M1 for 52×2.15 or 111.8 seen

M1 for their 111.8×0.84

93.91 **[A1]**

[Back to Question] *[Total Marks: 6]*

13 **(a)** Fuel used $= \frac{325}{100} \times 8.4$ or 27.3 **[M1]**

27.3×1.35 **[M1]**

36.855 or 36.86 **[A1]**

(b) 24 minutes = 0.4 hours or $\frac{2}{5}$ hours **[B1]**

Distance $= 75 \times 4.4$ **[M1]**

330 **[A1]**

[Back to Question] *[Total Marks: 6]*

14 **(a)** Volume in $m^3 = 1.8 \times 1.5 \times 1.2 = 3.24$ **[M1]**

$3.24 \times 1000 = 3240$ litres **[A1]**

(b) Time in seconds $= \frac{3240}{2.5} = 1296$ **[M1]**

Time in minutes $= \frac{1296}{60} = 21.6$ **[A1]**

(c) Rate $= \frac{\text{Volume}}{\text{Time}} = \frac{3240}{45}$ **[M1]**

$x = 72$ **[A1]**

[Back to Question] *[Total Marks: 6]*

15 **(a)** Volume $= 2.5 \times 1.4 \times 1.2$ or 4.2 (m^3) **[M1]**

Volume in litres $= 4.2 \times 1000$ or 4200 **[M1]**

Time in seconds $= \frac{4200}{0.35}$ or 12000 **[M1]**

3 h 20 min **[A1]**

(b) $\frac{780}{0.84}$ or 928.57... **[M1]**

$928.57 - 850$ **[M1]**

78.57 **[A1]**

[Back to Question] *[Total Marks: 7]*

16 **(a)** Time A to B $= \frac{d}{v}$ **[M1]**

Time B to C $= \frac{2d}{v+15}$ **[M1]**

Total Time $T = \frac{d}{v} + \frac{2d}{v+15} = \frac{d(v+15)+2dv}{v(v+15)} = \frac{dv+15d+2dv}{v(v+15)} = \frac{3dv+15d}{v(v+15)}$ **[A1]**

(b) Substitute $d = 120$ and $T = 4.5$ into the equation from part (a): $4.5 = \frac{3(120)v + 15(120)}{v(v+15)}$ **[M1]**

$4.5(v^2 + 15v) = 360v + 1800$ **[M1]**

$4.5v^2 + 67.5v - 360v - 1800 = 0$ **[M1]**

$4.5v^2 - 292.5v - 1800 = 0 \implies divideby1.5toget3\mathrm{v}^2 - 195v - 1200 = 0$ **[A1]**

[Back to Question] *[Total Marks: 7]*

17 **(a)** $\frac{x}{12} + \frac{x+5}{15} = \frac{31}{12}$ or $2\frac{35}{60}$ [M2]
(M1 for either $\frac{x}{12}$ or $\frac{x+5}{15}$ seen as a time)

$\frac{5x + 4(x+5)}{60} = \frac{155}{60}$ or equivalent correct clearing of fractions **[M1]**

$x = 15$ **[A1]**

(b) Total distance $= 15 + (15 + 5) = 35$ **[B1]**

Average speed $= \frac{35}{\frac{31}{12}}$ **[M1]**

13.5 **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** $\frac{2400}{v}$ **[B1]**

(b) $\frac{2400}{v} + \frac{2400}{v+40} = 10.5$ **[M1]**

$2400(v + 40) + 2400v = 10.5v(v + 40)$ **[M1]**

$2400v + 96000 + 2400v = 10.5v^2 + 420v$ **[M1]**

$10.5v^2 - 4380v - 96000 = 0$ **[A1]**

Multiply by $\frac{2}{3}$ to get $7v^2 - 2920v - 192000 = 0$ **[A1]**

(c) $v = \frac{-(-2920) \pm \sqrt{(-2920)^2 - 4(7)(-192000)}}{2(7)}$ **[M1]**

$v = 475.25...$ or $v = -57.68...$ (discard negative) **[A1]**

475 (correct to nearest integer) **[A1]**

[Back to Question] *[Total Marks: 7]*

19 **(a)** $\dfrac{30}{x} + \dfrac{18}{x-4} = 2.5$ **[M1]**

$\dfrac{30(x-4)+18x}{x(x-4)} = 2.5$ **[M1]**

$48x - 120 = 2.5(x^2 - 4x)$ **[M1]**

$48x - 120 = 2.5x^2 - 10x \implies 2.5x^2 - 58x + 120 = 0 \implies 5x^2 - 116x + 240 = 0$ **[A1]**

(b) $x = \dfrac{-(-116) \pm \sqrt{(-116)^2 - 4(5)(240)}}{2(5)}$ **[M1]**

$x = \dfrac{116 \pm \sqrt{13456 - 4800}}{10} = \dfrac{116 \pm \sqrt{8656}}{10}$ **[M1]**

$x = 20.90$ or $x = 2.30$ **[A1]**

[Back to Question] *[Total Marks: 7]*

20 **(a)** $\dfrac{15}{x} + \dfrac{12}{x-4} = 3.5$ **[M1]**

Multiply by $2x(x-4)$ to clear denominators (or equivalent): $30(x-4) + 24x = 7x(x-4)$ **[M1]**

$54x - 120 = 7x^2 - 28x \implies 7x^2 - 82x + 120 = 0$ **[A1]**

(b) **(i)** $(7x - 12)(x - 10) = 0$ OR $\dfrac{82 \pm \sqrt{(-82)^2 - 4(7)(120)}}{14}$ **[M2]**

If M0, allow M1 for $(7x \pm a)(x \pm b) = 0$ where $ab = 120$ or $7b \pm a = -82$

$x = 10$, $x = \dfrac{12}{7}$ (or 1.71) **[A1]**

(ii) Identifies valid speed $x = 10$ (since $x - 4 > 0 \implies x > 4$) **[B1]**

Time $= \dfrac{15}{10} = 1.5$ **[B1]**

[Back to Question] *[Total Marks: 8]*

21 **(a)** $\dfrac{150}{x}$ **[B1]**

(b) $\dfrac{150}{x} + \dfrac{90}{x+15} = 3$ **[M1]**

$\dfrac{150(x+15)+90x}{x(x+15)} = 3$ or $150(x+15) + 90x = 3x(x+15)$ **[M1]**

$150x + 2250 + 90x = 3x^2 + 45x$ **[M1]**

$3x^2 - 195x - 2250 = 0$ leading to $x^2 - 65x - 750 = 0$ **[A1]**

(c) $(x-75)(x+10)=0$ or $\dfrac{65 \pm \sqrt{(-65)^2-4(1)(-750)}}{2}$ **[M2]**

If M0 scored, allow [M1] for $(x \pm 75)(x \pm 10)=0$ or for correct substitution into the quadratic formula.

75 **[A1]**

[Back to Question] *[Total Marks: 8]*